CFO Payroll Intelligence Dashboard

Week 1 – SQL Queries & KPI Framework

Role: Data Science
Project: Executive Intelligence Foundation

Business Objective

The objective of this project is to design a CFO-level Payroll Dashboard using optimized SQL queries and aggregated logic to analyse payroll expenses, departmental cost distribution, employee cost metrics, and overtime impact.

The dashboard is built using three datasets:

Employees Table

Payroll Table

Attendance Table

Week 1 – SQL Queries & KPI Framework

1. Core Payroll KPIs

Total Payroll Cost

```
SELECT SUM(net_salary) AS total_payroll_cost
FROM payroll;
```

Total Gross Payroll

```
SELECT SUM(gross_salary) AS total_gross_salary
FROM payroll;
```

Total Tax Deduction

```
SELECT SUM(tax_deduction) AS total_tax
FROM payroll;
```

Total PF Deduction

```
SELECT SUM(pf_deduction) AS total_pf
FROM payroll;
```

Cost Per Employee

```
SELECT
SUM(net_salary) / COUNT(DISTINCT employee_id)
AS cost_per_employee
FROM payroll;
```

Department Expense

```
SELECT
e.department,
SUM(p.net_salary) AS department_expense
FROM payroll p
JOIN employees e
ON p.employee_id = e.employee_id
GROUP BY e.department;
```

Department Cost Per Employee

```
SELECT
e.department,
SUM(p.net_salary) / COUNT(DISTINCT p.employee_id)
AS dept_cost_per_employee
FROM payroll p
JOIN employees e
ON p.employee_id = e.employee_id
GROUP BY e.department;
```

Overtime Percentage

```
SELECT
(SUM(overtime_hours) / SUM(working_days)) * 100
AS overtime_percentage
FROM attendance;
```

Absenteeism Rate

```
SELECT
((SUM(working_days) - SUM(present_days)) / SUM(working_days)) * 100
AS absenteeism_rate
FROM attendance;
```

Headcount by Department

```
SELECT
department,
COUNT(employee_id) AS total_employees
FROM employees
GROUP BY department;
```

2. Aggregated Views for Dashboard Optimization

CFO Summary View

```
CREATE VIEW vw_cfo_summary AS
SELECT
    SUM(net_salary) AS total_payroll,
    SUM(gross_salary) AS total_gross_salary,
    SUM(tax_deduction) AS total_tax_deduction,
    SUM(pf_deduction) AS total_pf_deduction,
    COUNT(DISTINCT employee_id) AS total_employees
FROM payroll;
```

Department Expense View

```
CREATE VIEW vw_department_expense AS
SELECT
    e.department,
    SUM(p.net_salary) AS department_expense
FROM payroll p
JOIN employees e
ON p.employee_id = e.employee_id
GROUP BY e.department;
```

Attendance Summary View

```
CREATE VIEW vw_attendance_summary AS
SELECT
    SUM(working_days) AS total_working_days,
    SUM(present_days) AS total_present_days,
    SUM(overtime_hours) AS total_overtime_hours
FROM attendance;
```

3. Conclusion

The SQL queries were designed to:

- Provide CFO-level cost visibility
- Enable department-level expense analysis
- Identify overtime impact
- Analyse payroll distribution
- Support trend-based financial insights

The dashboard integrates aggregated SQL logic with business-focused KPIs to deliver executive-ready intelligence.